Experimental Design

Pietro Franceschi | pietro.franceschi@fmach.it

Statistics

Statistics is the science which deals with **the collection, the analysis, the visualization and the interpretation** of experimental data.

How data are collected ...

- Random samplings
- Observational studies
- Experiments



Random Samplings (polls)

Definition

Random samplings allow to characterize the properties of a finite population without measuring all of its members.

Examples

- *Electoral polls*
- *Normal levels of cholesterol in the human population*
- *Characterization of a population of grapes, apples, wines, ...*

Observational Studies

Definition

Observational studies are designed with the objective of **identifying relationships** between the different properties of a conceptual population. The role of the experimenter is to perform the **selection of the sample**.

Examples

- *Is it true that people who eat more chocolate are more happy?*
- *The level of cholesterol of people eating more vegetables is lower*

Experiments

Definition

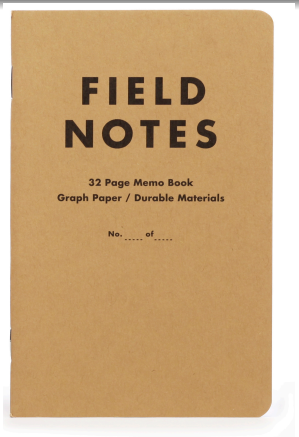
Experiments are designed with the objective of identifying **causal relations** between the properties of a conceptual population. The role of the experimenter is to **modify the conditions** to verify the presence of causal relationship between the observed properties.

Examples

- *If you eat more chocolate you will get happier*
- *If I drink more beer I'll get more sympathetic*

Important notes

- Causal relations can be assessed only in **experiments**
- This is really Galileian ;-)
- Experiments are impossible in many relevant fields like human health and ecology



Should we then give up on obtaining causal information there?

Mind the chocolate ...

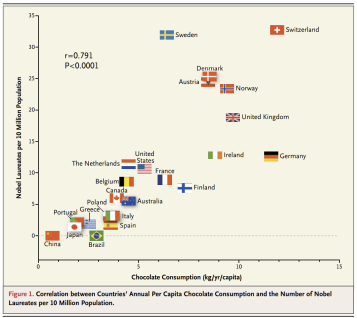


Figure 1. Correlation between Countries' Annual Per Capita Chocolate Consumption and the Number of Nobel Laureates per 10 Million Population.

New England Journal of Medicine, 2012

" ... Chocolate consumption enhances cognitive function, which is a sine qua non for winning the Nobel Prize, and it closely correlates with the number of Nobel laureates in each country ..." **...and more!**

Planning a sampling

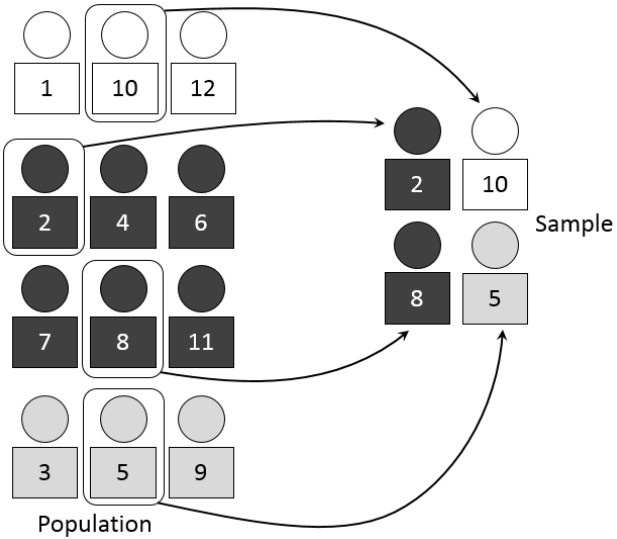
Key question

What is the best way to sample my population in a representative way?

- Do it randomly to avoid any intentional or unintentional bias (**Randomized Sampling**)
- Take into account known subpopulations and confounding factors (**Stratified Random Sampling**)
- The number of samples is determined by practical/economical considerations

In presence of known subpopulations, stratified random sampling results in a more accurate characterization of the population

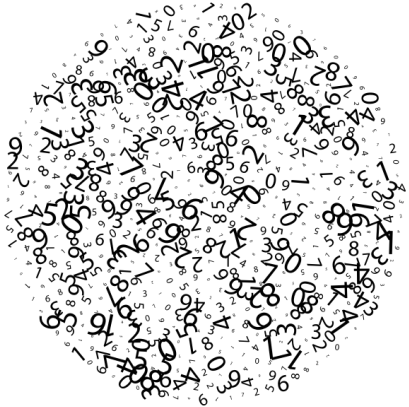
Stratified Random Sampling



Key idea: do it randomly

The most reasonable way to "smear out" the effects of unknown biases is to do everything **randomly**

Random is not a synonym of HAPHAZARD



Planning an observational study

Objective

get a useful and clear answer.
Means: start from a clear, useful and often simple question.

- Identify the **sampling unit**
- Decide the number of samples (money, power, ...)
- Define the conceptual population
- Sample it in a representative way
- Identify **confounding factors** and, if possible, **stratify** for them

Key idea: Sampling Unit

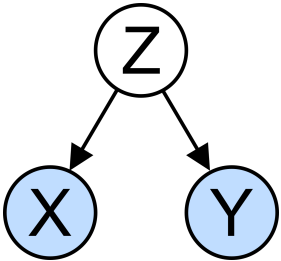
The smaller unit of a population which retains the properties we are interested in.

- *Example: grapevine, leafs, infections ...*

Key idea: Confounding Factor

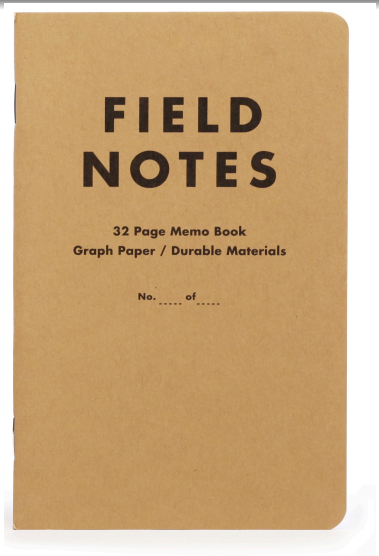
A variable that influences both the dependent variable and independent variable, causing a spurious association (*wikipedia*).

- *Smoke, cardiovascular disease, alcohol consumption*
- *Birth order (1st child, 2nd child, etc.), maternal age, Down Syndrome in the child*



Notes

- Some confounders can be controlled by careful sampling
  - *Age and Gender on the relation between happiness and chocolate*
- Some others are impossible to control
  - *Presence of chemical pollution in water streams and altitude of sampling*



Planning one experiment

Objective

get a useful and clear answer.
Means: start from a clear, useful and often simple question.

- What is my **experimental unit**?
- How many samples should I measure?
- What are the **potential sources of variability**?

Experimental Design

Definition

A strategy to assign the experimental units to the different *treatments* to optimize my capacity of inferring **causal relationships**.

Control of unwanted sources of variability (technical/biological) to highlight the effects of the intervention

The tools: **randomization** and **blocking**

Key tool: Blocking

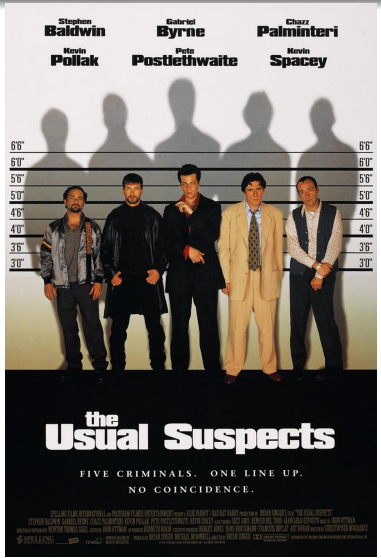
- Group experimental units in homogeneous groups (**blocks**)
- Perform a "full" experiment within each block (blocks and study factors will be **orthogonal**)
- Identify and subtract the variability across the blocks



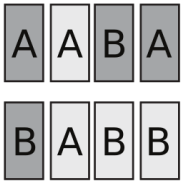
Block what you can; randomize what you cannot ...

Examples of common blocking factors

- Location
- Analytical batch
- Day
- Operator



Complete Randomized Design (CRD)

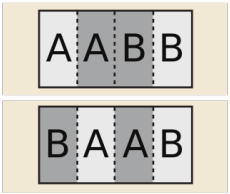


Irrigation:

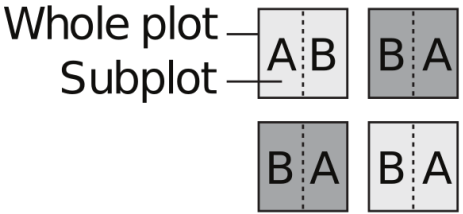
Fertilizer: A,B

Field (block):

Randomized Complete Block Design (CRD)



Split Plot + CRD

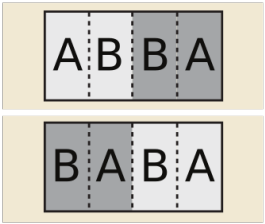


Irrigation:

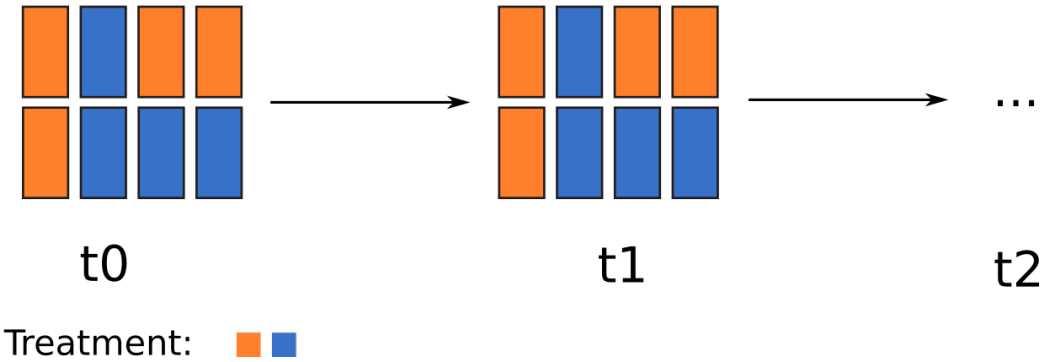
Fertilizer: A,B

Field (block):

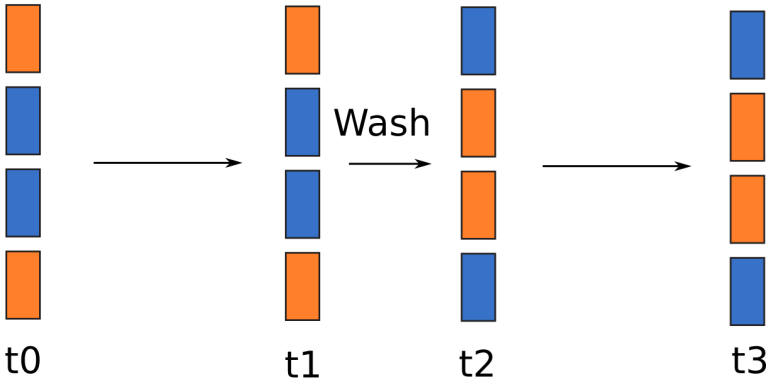
Split Plot + RCBD



Longitudinal (repeated measures)



Cross-over (repeated measures)



Notes

- Block as much as possible!
- Repeated measures are more "powerful" because each unit is the control of itself
- Crossovers can be tricky for the wash-out
- Repeated measures designs are key when population variability is large (e.g. plants in the field/greenhouse)

